Sahaj Bhadja

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Summary

Frontend product engineer with 2+ years shipping **TypeScript/React interfaces**, **AI-enabled workflows**, and **API-driven tools** from prototype to production with fast user-feedback iteration and strong product ownership.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built **React + Flask interfaces** for 10K+ daily requests, turning ambiguous user requirements into operator-friendly workflows for grading, feedback, and model-guided decision support across production academic operations.
- Shipped rapidly using AI-assisted development and CI/CD validation, reducing turnaround by **60%** while continuously refining UX behavior through real user interactions and tight collaboration with product stakeholders.

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Developed **user-facing data interfaces** and backend services for 500+ users, reducing latency from **1.2s to 350ms** and improving workflow fluidity for high-frequency interaction patterns in production.
- Implemented FastAPI orchestration for 5K+ monthly jobs with validation and observability hooks, enabling faster interface iteration cycles and higher confidence when deploying AI-adjacent operational features.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **MERN application modules** with REST APIs and WebSockets supporting eight workflows and 2M+ synchronized records, improving real-time interface responsiveness and maintaining consistent user state across sessions.
- Automated release workflows with Docker and Jenkins, reducing deployment time by **50%** and enabling faster product experiments, bug-fix loops, and confident rollout of user-visible improvements.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered API and ETL services with Python/Java and AWS Glue/S3, improving response times by **30%** and ensuring front-end workflows remained reliable during high-volume backend processing.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, improving CI efficiency by **68%** and reducing technical debt by **20%** through robust quality controls.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built Python evaluation pipelines over 100K+ trajectories and summarized findings for engineers and researchers, improving trust in AI behavior analysis and accelerating decisions on feature-level improvements.
- Partnered across disciplines to convert experimental insights into actionable product-facing changes, strengthening feedback loops between model behavior, interface design choices, and operational reliability expectations.

Skills

Frontend: **TypeScript**, **React.js**, component libraries, UX workflow design, API-driven interfaces
AI Product Workflows: LLM-integrated flows, evaluation/feedback loops, AI-assisted development tooling
Backend: Python, Flask, FastAPI, Node.js, REST APIs, asynchronous orchestration
Delivery: CI/CD, Git, Docker, Jenkins, production debugging, rapid prototype-to-production iteration

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **real-time product interface** for 50K+ users and 1M+ transactions with secure API integrations, improving workflow clarity and enabling high-velocity iteration on financial interaction patterns.

PiXMod | *Next.js, Python, MongoDB, Transformers*

- Developed an AI-native full-stack SaaS experience with segmentation and retrieval workflows, processing 1K+ assets while refining interface controls to improve user trust, speed, and operational throughput.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a schema-aware query interface across 10+ databases with structured outputs and validation, enabling users to guide AI-assisted analysis workflows and iterate rapidly from prompt to trustworthy results.